

Keywords: topological reasoning • spatial cognition • multimodal LLM • benchmark • knot theory

MindTopo: Can Foundation Models Reason in Topological Space?

Best Model Scores 61.4% vs. 97.9% Human on a Benchmark of 11,030 Topological Intuition Instances

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arXiv:2609.11900

Spatial reasoning benchmarks for foundation models have focused almost entirely on metric properties: distances, angles, viewpoints, and shapes. Cognitive science identifies a more fundamental layer of spatial understanding — topological relations that remain invariant under continuous deformation — as the earliest spatial concepts acquired by infants. MindTopo evaluates foundation models on this layer and finds a systematic blind spot: the best multimodal LLM scores 61.4% on 11,030 topological instances against 97.9% human performance, with near-random performance on knots — the most globally structured topological property.

AT A GLANCE

Problem: Foundation models are not evaluated on topological spatial relations despite their cognitive primacy.

Why it matters: Topological reasoning underlies planning, navigation, and physical interaction; a gap here limits physical AI.

Method: 11,030 procedurally generated instances across 5 topological properties (continuity, separation, order, enclosure, knots) at reasoning and planning cognitive levels.

Result: Best model 61.4%; human 97.9%; knots near random for all 14 tested multimodal LLMs.

Evidence: 14 MLLMs + 3 video generative models evaluated; ablations with chain-of-thought, metric cue removal, difficulty scaling.

Topology vs. Metric Geometry

Topology studies properties invariant under *homeomorphisms* (continuous deformations with continuous inverses), as opposed to geometry, which studies properties

invariant under rigid motions. The classic example: a coffee cup and a donut are topologically equivalent (both have one handle/hole) but geometrically different.

Cognitive developmental science (Piaget 1956; Spelke et al. 1995) identifies topological relations as the first spatial concepts acquired by infants, preceding projective and metric relations. Despite this cognitive primacy, virtually all spatial reasoning benchmarks for foundation models measure metric or viewpoint-dependent properties.

MindTopo addresses this gap with five topological properties:

Continuity: Is a path or region connected without breaks?

Separation: Are two regions or paths disconnected?

Order: What is the cyclic ordering of points along a closed curve?

Enclosure: Does one region contain another (Jordan curve-type)?

Knots: What is the global entanglement class of a closed curve in 3D space?

The Knot Problem

Knots are the hardest property in MindTopo and the most revealing failure. A *knot* is a closed curve $\gamma : S^1 \rightarrow \mathbb{R}^3$. Two knots are equivalent (the same knot type) iff one can be deformed into the other without cutting — a relation that cannot be decomposed into any set of local features.

By the Reidemeister theorem, two knot diagrams represent equivalent knots iff one can be obtained from the other by three local moves (Reidemeister moves I, II, III). Checking equivalence requires reasoning about global path structure through an unbounded sequence of local moves. Current vision models, which process local patches and aggregate them, have no mechanism for this kind of global topological bookkeeping.

Result: All 14 MLLMs score between 24–35% on knot instances (4-choice random = 25%), confirming the problem is currently unsolvable by the tested models regardless of scale or architecture.

Benchmark Design

MindTopo contains **11,030 instances** across **13 task types** at two cognitive levels:

Reasoning: Given visual input (image or short video), identify a topological property or predict how it changes under a described deformation.

Planning: The model acts as a closed-loop agent selecting environment actions. Task completion within $T = 20$ steps is the primary metric.

Instances are generated algorithmically using a custom topology simulator with difficulty parameter $d \in [1, 5]$ controlling the number of topological operations required. All instances are procedurally generated — not scraped — ensuring no overlap with web training data.

Human baseline: 50 participants (college students with no topology training) evaluated a stratified 500-instance subset; mean accuracy 97.9%.

Technical Formulation

Let $f : X \rightarrow Y$ be a continuous map between topological spaces. MindTopo operationalizes five invariant types:

Separation: Two sets $A, B \subset X$ are separated if $\bar{A} \cap B = A \cap \bar{B} = \emptyset$.

Enclosure: Region A encloses B if $B \subset \text{Int}(A)$ in the ambient space (Jordan curve theorem generalization).

Knot equivalence: Closed curve $\gamma : S^1 \rightarrow \mathbb{R}^3$ belongs to knot class $[K]$ if it is ambient isotopic to K . MindTopo uses three-crossing and four-crossing knots as test cases.

Order invariant: Given n points on a closed curve, their cyclic order $\sigma \in S_n / \mathbb{Z}_n$ is invariant under homeomorphism.

Experimental Findings

Model	Reasoning	Planning
Human	98.2%	97.5%
Gemini 1.5 Pro (best)	68.3%	54.5%
GPT-4o	65.1%	48.9%
Claude 3.5 Sonnet	63.7%	46.2%
Avg. 14 MLLMs	52.4%	38.7%

Results by property reveal a clear hierarchy:

Property	Best Model	Human
Continuity	79.2%	98.8%
Separation	74.1%	98.5%
Order	65.8%	97.4%
Enclosure	61.3%	97.1%
Knots	29.4%	97.2%

Continuity and separation (local, visually apparent properties) are relatively easier; knots (global, non-local) are near-random.

Ablations and Interpretation

Metric cues removed: When images are topologically equivalent but metrically different (differently scaled loops), model accuracy changes by less than 2%, confirming models do not exploit metric shortcuts.

Chain-of-thought prompting: CoT improves average

accuracy by 4.1 points on reasoning and 3.8 on planning — helpful but insufficient to close the human gap.

Difficulty scaling: At difficulty $d = 5$, models average 31.2% (near random) while humans score 95.4%, indicating complex topological configurations are qualitatively out of distribution for current models.

Planning gap: All models perform better on reasoning than planning (average gap: 13.7 points), suggesting models recognize topological cues without grounding them in action-relevant planning representations.

Reference

Yunfei Ge et al. **MindTopo: Can Foundation Models Reason in Topological Space?** arXiv:2609.11900, September 2026. <https://arxiv.org/abs/2609.11900>